Thermodynamic characterization of the boundary layer under fog and dew events in the coastal hyper-arid climate of the Atacama Desert

F. Muñoz [1] , F. Lobos-Roco[1,2], S. Acevedo[1], C. Del Río[2]

1 Facultad de Agronomía y Sistemas Naturales, Pontifica Universidad Católica de Chile
2 Centro UC Desierto de Atacama

FACULTAD DE AGRONOMÍA Y SISTEMAS NATURALES
PONTIFICIA UNIVERSIDAD CATÓLICA DE CHILE

Centro UC Desierto de Atacama

1. MOTIVATION AND PROBLEM

Atm. water = Precipitation (negligible) + Fog + Dew

- Fog and dew are key water sources in hyper-arid environments.
- Standard collectors (SFC/SDC) cannot reliably distinguish between them (Fig 1., instrumental collection).
- Clear separation is critical for accurate water balance estimates and water harvesting design.
- Despite their importance, dew remains significantly less studied than fog.



Fig. 1. Fog in Atacama desert.



Fig. 2. *Tillandsia's fields.*

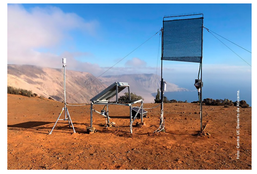


Fig. 3. Fog and dew collectors (SFC and SDC)

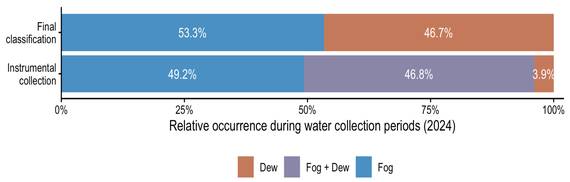


Fig. 4. Event proportions at station 1211, cerro Oyarbide.

2. THEORETICAL FRAMEWORK

- Both processes occur under near-saturated atmospheric conditions.
- Fog is associated with turbulent boundary layers.
- Dew forms under stable atmospheric conditions.
- Potential temperature and specific humidity are used to characterize these regimes.

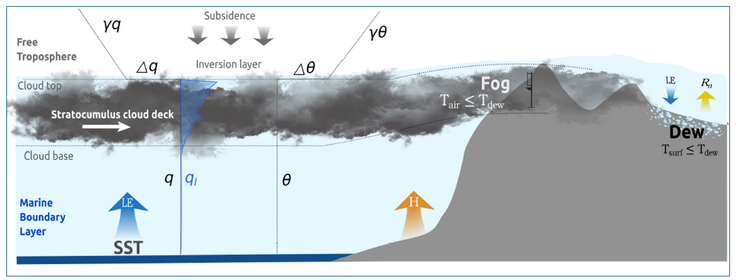
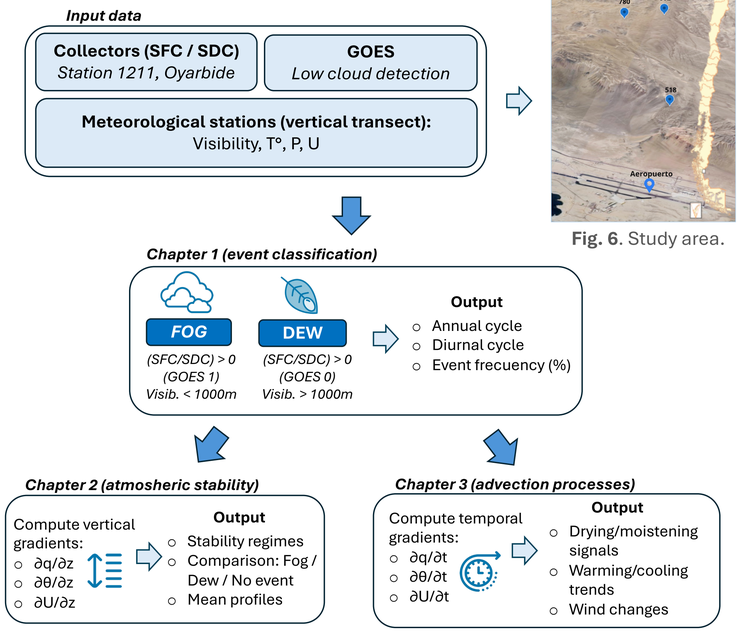


Fig. 5. Schematic representation of fog and dew formation processes.

3. METHODOLOGICAL FLOW



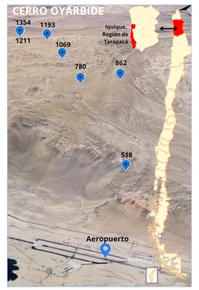


Fig. 6. Study area.

4. RESULTS AND DISCUSSION

CHAPTER 01: CLASSIFICATION

- In Fig 4., most events are classified as dew, suggesting a larger role than previously assumed.
- In Fig 7., Fog and dew show opposite annual frequency patterns, with peak collection in July–October.
- In Fig 8., Water collection peaks at night, with an inverse relation to event frequency.

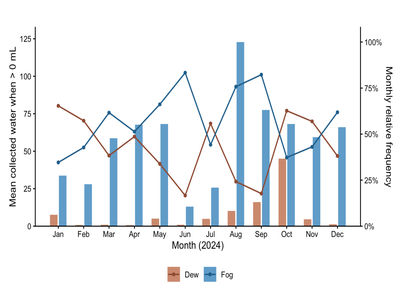


Fig. 7. Annual variability of fog and dew.

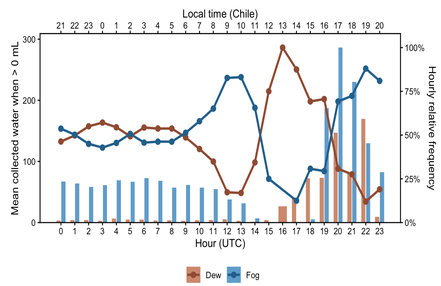


Fig. 8. Diurnal cycle of fog and dew.

CHAPTER 02: ATMOSPHERIC STABILITY

- Vertical gradients of q, θ, and U are analyzed for events identified at 1211 m.
- Δq/Δz is similar across all conditions, consistent with near-saturation in fog and dew.
- Δθ/Δz is higher for dew, slightly exceeding the fog threshold (0.0031), indicating more stable conditions.
- ΔU/Δz shows no clear differences between events.

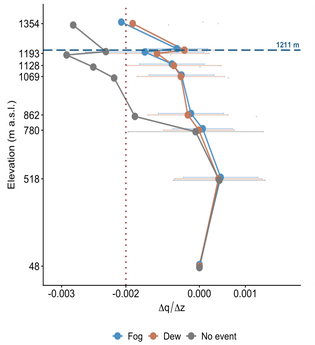


Fig 9. Vertical gradiente of specific humidity.

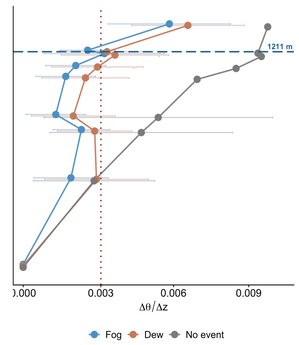


Fig 10. Vertical gradiente of potential temperature.

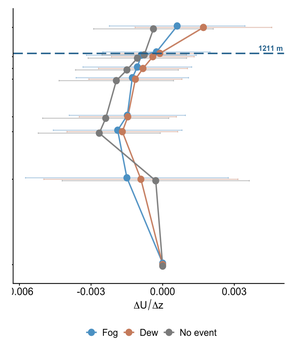


Fig 11. Vertical gradiente of wind speed.

CHAPTER 03: TEMPORAL EVOLUTION

- Fog shows increasing Δq and decreasing ΔU, consistent with moist advection and reduced turbulence.
- Dew shows stable Δq and higher Δθ, indicating formation under radiative cooling and stable conditions.

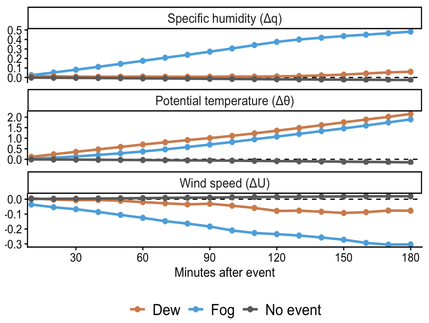


Fig 12. Post-event evolution.

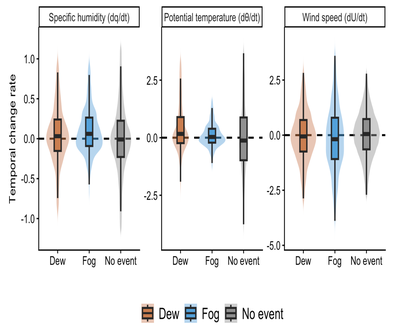


Fig 13. Temporal gradients.

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